AI-POWERED PERSONAL FINANCE TRACKER

NIIT Capstone Project - III | Data Analytics & Automation using Python

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What is the Finance Tracker?

The AI-Powered Personal Finance Tracker is a Python-based application that helps individuals keep their income and expenses organised in one place. It calculates the net balances, shows spending patterns across categories and provides an AI assistant that can answer questions using the financial information entered by the user.

How does it work?

The application is structured into a logical, three-stage workflow that takes users from basic record-keeping to advanced financial analysis and AI-driven intelligence:

1. **Stage 1: Record, save, and manage transactions:**
   Easily log income or expenses with a unique ID, category, amount, and date. The app keeps entries clean by catching typos, duplicate IDs, or invalid numbers, while automatically syncing everything to an Excel spreadsheet (finance_data_final.xlsx) so your data is never lost.
   Stage 2: Understand spending and analyze performance:
   Instantly view your net balance, total income, and total expenses. Using Pandas and NumPy, the app groups spending by category to pinpoint big cost drivers (like Rent or Bills), flags large expenses over ₹2,000, and tags your habit as a Saver, Balanced, or Spender.
2. **Stage 3: AI Financial Assistant (Groq LLM API):**
   Powered by the Groq LLM API, the AI assistant translates your numbers into clear, practical guidance. It offers two main features:
   (1) automated standard reports summarizing your top expenses and budget tips, and
   (2) a conversational chat mode where you can ask custom questions to get personalized financial advice.

Key benefits

- Keeps income and expenses organised.
- Makes balances and spending patterns easier to understand.
- Highlights large expenses and the categories that use the most money.
- Keeps records accessible through Excel import and export.
- Offers AI-based responses to users' finance questions.

What the user needs

- Keep the Excel file available for saving or restoring records. (finance_data_final.xlsx)
- The AI question-and-answer feature requires a working internet connection and a configured Groq API